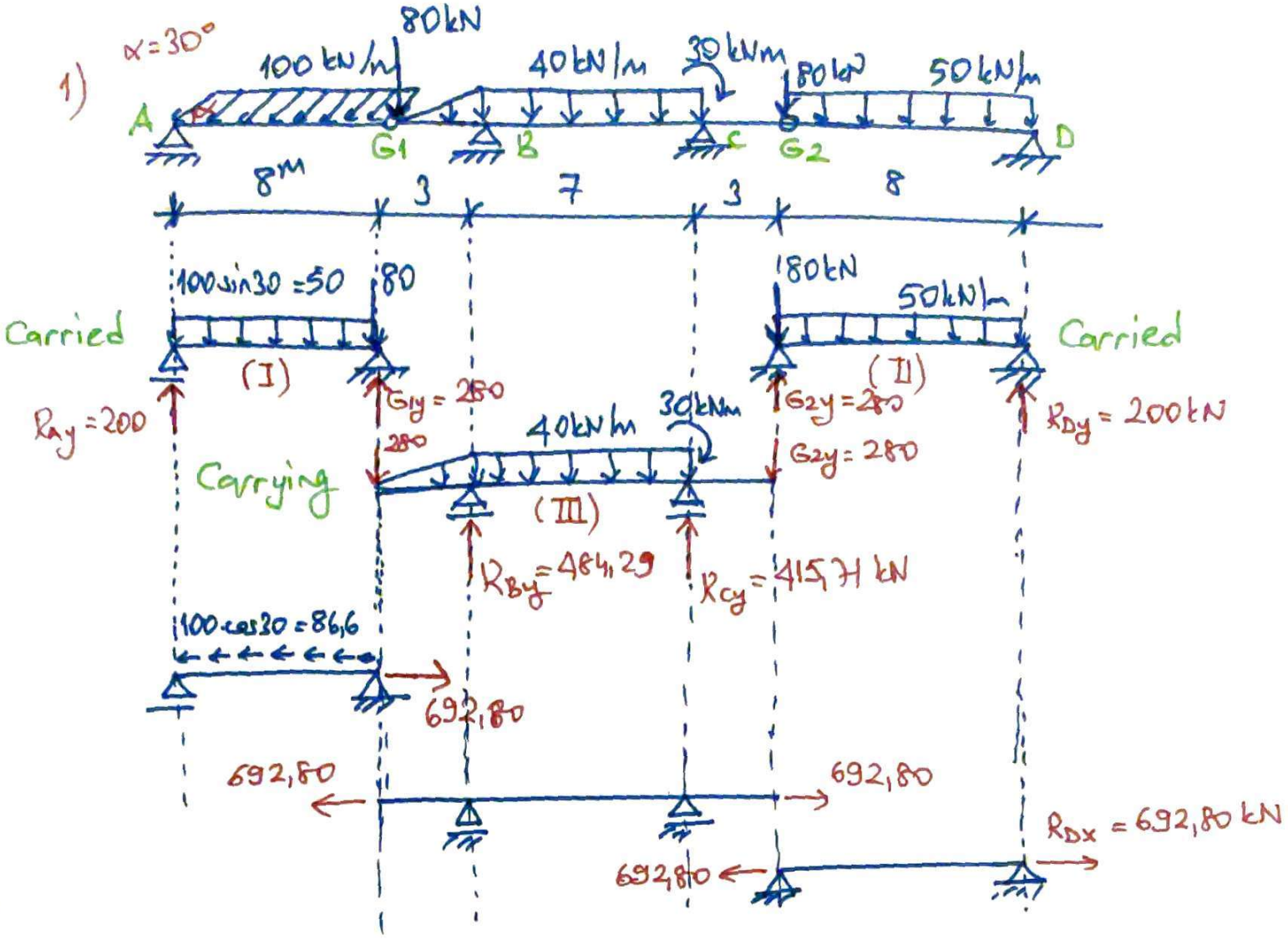
- GERBER BEAMS EXAMPLES -

1)



(I) and (II) parts:

$\Sigma M_{G1} = 0 \Rightarrow R_{Ay} = \frac{50.8.4}{8} = 200\,kN$

$G_{iy} = \frac{50.8.4 + 80.8}{8} = 280\,kN$

$V_{AG1} = 200\,kN \quad V_{G1A} = -280 + 80 = -200\,kN$

$x_0 = \frac{200}{50} = 4\,m \Rightarrow M_{max} = 200.4 - 50.4.2 = 400\,kNm$

Part (III):

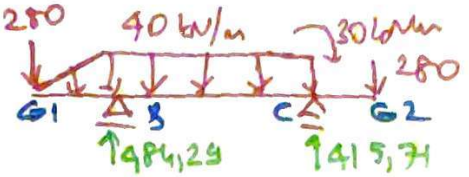
$\Sigma M_C = 0 \Rightarrow R_{By} = \frac{280.10 + \frac{40.3}{2}.(7+1) + 40.7.3,5 - 30 - 280.3}{7}$

$R_{By} = 484,29\,kN$

$\Sigma M_B = 0 \Rightarrow R_{Cy} = \frac{280.10 + 30 + 40.7.3,5 - \frac{40}{2}3.1 - 280.3}{7}$

$R_{Cy} = 415,71\,kN$

Internal Loads of part III:



G1 - B

$V_{G1B} = -280\ kN \qquad V_{BG1} = -280 - \frac{40 \cdot 3}{2} = -340\ kN$

$M_B = -280 \cdot 3 - 40 \cdot \frac{3}{2} \cdot 1 = -900\ kNm$

B C

$V_{BC} = 484,29 - 280 - \frac{40 \cdot 3}{2} = 144,29\ kN$

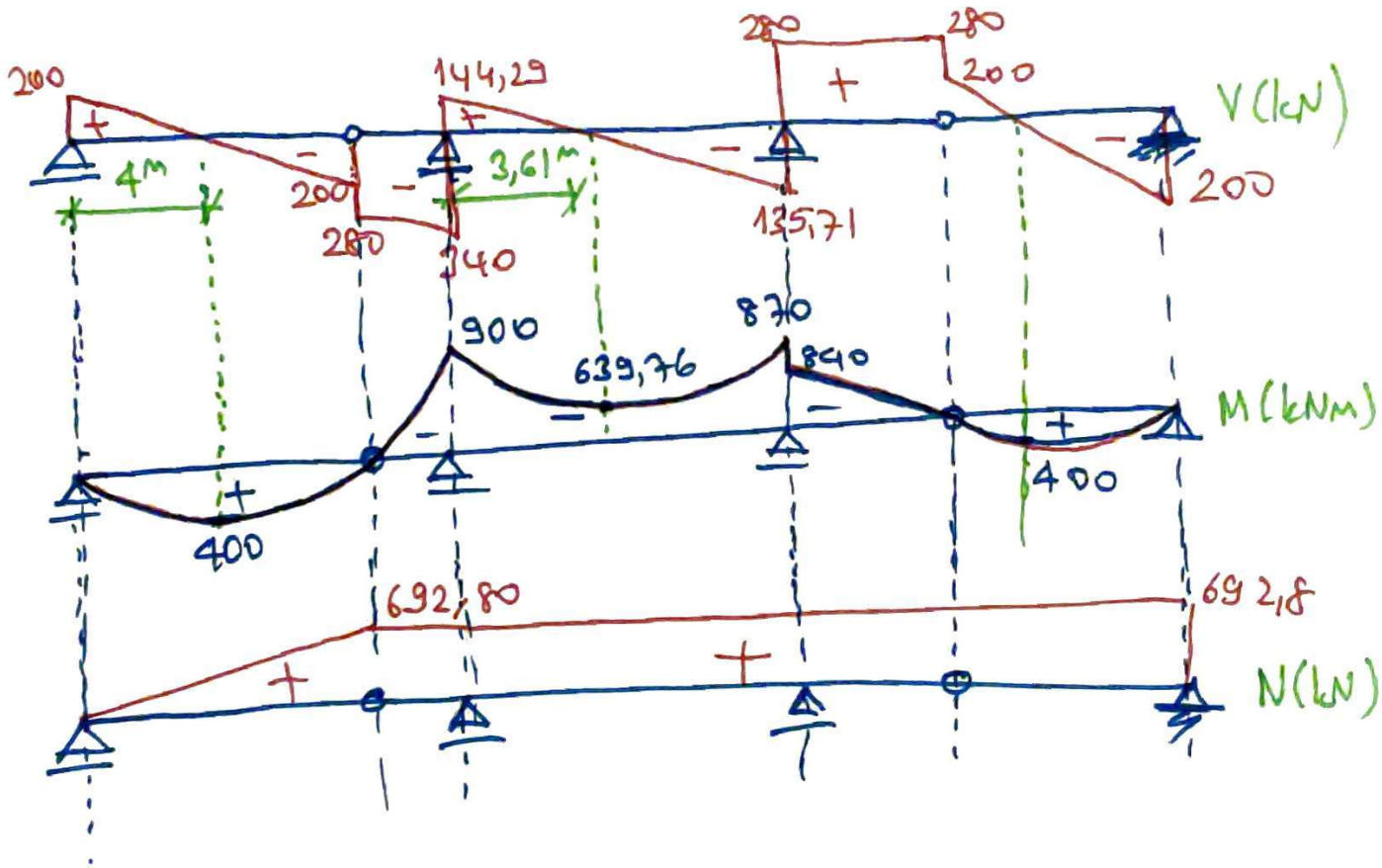
$V_{CB} = -415,71 + 280 = -135,71\ kN$

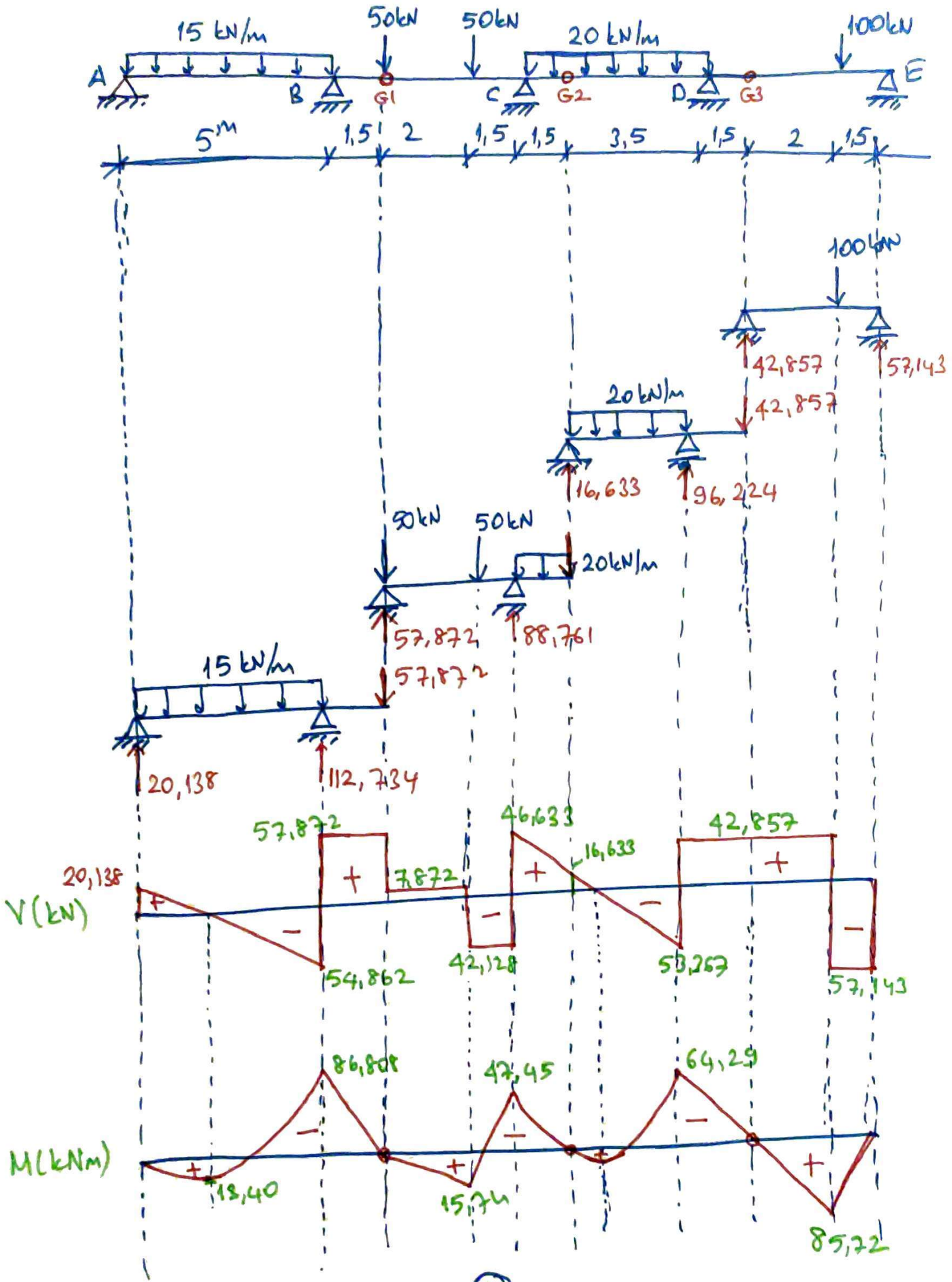
$M_{CG2} = -280 \cdot 3 = -840\ kNm$

$M_{CB} = -280 \cdot 3 - 30 = -870\ kNm$

$484,29 - 280 - \frac{40 \cdot 3}{2} - 40 \cdot x_0 = 0 \Rightarrow x_0 = 3,61\ m$

$M_{max} = 484,29 \cdot 3,61 - 280 \cdot 6,61 - \frac{40 \cdot 3}{2} \cdot (1 + 3,61) = -639,76\ kNm$




15 kN/m
50kN
50kN
20 kN/m
100kN
A
B
G1
C
G2
D
G3
E
5m
1,5
2
1,5
1,5
3,5
1,5
2
1,5
100kN
42,857
57,143
20 kN/m
42,857
16,633
96,224
50kN
50kN
20kN/m
57,872
88,761
15 kN/m
57,872
20,138
112,734
V(kN)
20,138
57,872
7,872
54,862
46,633
16,633
42,128
42,857
53,267
57,143
M(kNm)
18,40
86,808
15,74
47,45
64,29
85,72

- Structural Layout of Gerber Beams -

